$$\gamma : \underbrace{3x^2 \overset{2a_{12}}{-2\sqrt{3}}\,xy + y^2}_{} - 2x - 2\sqrt{3}y = 0$$

Caso $a_{12} \neq 0$

$\overset{\lambda_1\lambda_2}{\det A} = 0 \Rightarrow \gamma$ parabola

$$\sqcup = [x, y] \cdot A \cdot \begin{bmatrix} x \\ y \end{bmatrix} \quad \text{dove } A = \begin{bmatrix} 3 & -\sqrt{3} \\ -\sqrt{3} & 1 \end{bmatrix}$$

$$P(\lambda) = |A - \lambda I| = \begin{vmatrix} (3-\lambda) & -\sqrt{3} \\ -\sqrt{3} & (1-\lambda) \end{vmatrix} = \lambda^2 - 4\lambda + 3 - 3 = \lambda(\lambda - 4)$$

$\lambda_1 = 4 \quad \lambda_2 = 0$ Determiniamo rotazione P t.c.

$${}^tP \cdot A \cdot P = \begin{bmatrix} 4 & 0 \\ 0 & 0 \end{bmatrix} \qquad P = [\underset{\substack{\uparrow \\ \text{base o.n.} \\ \text{di } V_{\lambda_1}}}{\vec{w}_1}, \underset{\substack{\uparrow \\ V_{\lambda_2}}}{\vec{w}_2}]$$

$$A - \lambda_1 I = \begin{bmatrix} -1 & -\sqrt{3} \\ -\sqrt{3} & -3 \end{bmatrix}$$

Quindi $\vec{w}_1$ ricavo da $\underbrace{(A - \lambda_1 I)}_{\rho = 1}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$, cioè

$$\begin{cases} -x - \sqrt{3}y = 0 \\ \cancel{-\sqrt{3}x - 3y = 0} \end{cases}$$

$$\vec{w}_1 = \text{vers}\left(\begin{bmatrix} \sqrt{3} \\ -1 \end{bmatrix}\right) = \begin{bmatrix} \sqrt{3}/2 \\ -1/2 \end{bmatrix}$$

$$\vec{w}_2 \ \ldots \qquad \vec{w}_2 = \begin{bmatrix} 1/2 \\ \sqrt{3}/2 \end{bmatrix} \qquad P = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = P \begin{bmatrix} x' \\ y' \end{bmatrix} \iff \begin{cases} x = \frac{1}{2}(\sqrt{3}x' + y') \\ y = \frac{1}{2}(-x' + \sqrt{3}y') \end{cases}$$

Ora trasformiamo eq. di γ rispetto alle coord. x', y':

$$\gamma : \underbrace{3x^2 - 2\sqrt{3}xy + y^2}_{} - 2x - 2\sqrt{3}y = 0 \qquad \sqcup = \lambda_1 x'^2 + \lambda_2 y'^2$$

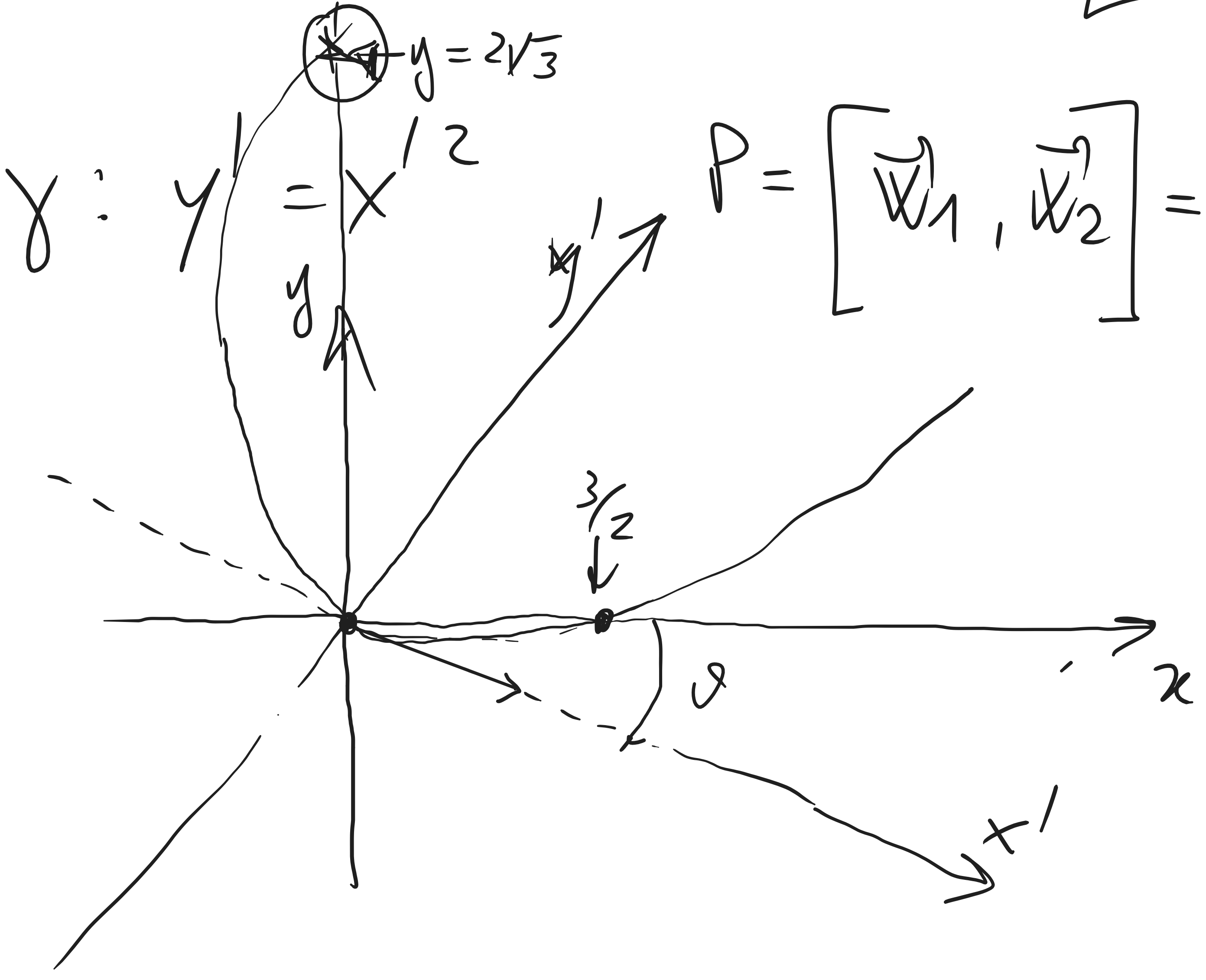
$$\gamma : \underbrace{4x'^2}_{} - (\sqrt{3}x' + y') + \sqrt{3}x' - 3y' = 0$$

$$\gamma : 4x'^2 - 4y' = 0 \qquad \boxed{y' = x'^2}$$

forma canonica,

Non è necessaria ulteriore traslaz.

$$\gamma : y' = x'^2 \qquad P = [\vec{w}_1, \vec{w}_2] = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$$



$$\vartheta = -\frac{\pi}{6}$$

$$\bullet \begin{cases} \gamma \\ y = 0 \end{cases} \quad 3x^2 - 2x = 0 \qquad x = \frac{3}{2} \quad x = 0$$

$$* \begin{cases} \gamma \\ x = 0 \end{cases} \quad y^2 - 2\sqrt{3}y = 0 \quad y = 2\sqrt{3}$$